Fixing weak language grounding in a robot policy's weights

Three robot-policy architectures, one released model of each: a VLA with an action expert (OpenPI π0.5), a VLA on a reasoning backbone (GR00T N1.7 DROID, Cosmos-Reason2-2B) and a world-action model (Cosmos3 Edge Policy), on the same RoboLab DROID tasks Cube then banana, Eyeglasses, Cube left of bowl, Yellow on red, Mustard in left bin, Three-cube tower, Flat box, Mustard right of raisin box and a retention task Cube in bowl.

Method. When a clearer wording of the same goal makes a robot succeed, copy that wording's internal effect into the weights: one least-squares calculation per layer, no training, and the robot still gets the original instruction.

VLA with an action expert (π0.5). The edit fixed its two grounding failures: Cube then banana 2 → 40 of 64 unseen starts and Mustard right of raisin box 0 → 28 of 64; a random edit of the same size did nothing (2 vs 2 of 64).

Across architectures. The fix: A closed-form weight edit makes a world-action model (Cosmos3 Edge) succeed from 56 of 64 new starts, as often as rewording the instruction does. **When it works:** Only in the VLM or text pathway. **The cost:** It breaks the two other tasks that say "left", and rebuilding it to protect them didn't help.



Cosmos3 Edge Policy, Mustard right of raisin box: the wording gap; the clip plays on the live page.

What the edit did on each task, on unseen starts unless marked practice

	Cosmos3 Edge Policy world-action model	**GR00T N1.7** reasoning-backbone VLA	**π0.5** OpenPI VLA, control
Cube then banana	Mustard right of raisin box edit: 18 → 18 of 32 9 / 9; p = 1	9 → 0 of 32 practice starts 0 / 9, p = 0.0039	2 → 40 of 64 38 / 0, p = 7.3×10^{-12}; paper's evaluation: fixed: 27 → 106 of 128
Eyeglasses	Mustard right of raisin box edit: 4 → 1 of 16 0 / 3; p = 0.25	unedited 0 of 32 not edited	Cube then banana edit: 12 → 21 of 32 12 / 3; p = 0.035; paper's evaluation: Cube then banana edit: 22 → 44 of 64
Cube left of bowl	Mustard right of raisin box edit: 13 → 0 of 32 0 / 13; p = 2.4×10^{-4}	unedited 1 of 32 not edited	Cube then banana edit: 1 → 6 of 32 6 / 1; p = 0.12; paper's evaluation: Cube then banana edit: 8 → 36 of 64
Yellow on red	Mustard right of raisin box edit: 24 → 22 of 32 4 / 6; p = 0.75	unedited 0 of 16 not edited	no gain lookalike perception; no clear wording
Mustard in left bin	Mustard right of raisin box edit: 9 → 0 of 32 0 / 9; p = 0.0039	unedited 0 of 32 not edited	no gain own edit 8 → 11 of 64; the grasp slips
Three-cube tower	Mustard right of raisin box edit: 1 → 1 of 32 1 / 1; p = 1	unedited 0 of 16 not edited	no gain own edit 35 → 39 of 64
Flat box	Mustard right of raisin box edit: 0 → 0 of 16 0 / 0; p = 1	unedited 0 of 32 not edited	no gain a missing skill; 0 of 64 unedited
Mustard right of raisin box	0 → 56 of 64 56 / 0, p = 2.8×10^{-17}	unedited 0 of 32 not edited	0 → 28 of 64 28 / 0, p = 7.5×10^{-9}; paper's evaluation: fixed: 0 → 27 of 64
Cube in bowl	24 → 28 of 32	unedited 28 of 32 not edited	31 → 31 of 32 paper's evaluation: kept: 32 → 32 of 32
Rendering: performance → default preset	—	5 → 16 of 16 Banana onto plate	—
Random control, same size	control 0, unedited 0, edit 56 of 64 Mustard right of raisin box	control 11, unedited 9, edit 0 of 32 practice starts Cube then banana	control 2, unedited 2, edit 40 of 64 Cube then banana
Where to edit	text pathway 13, video-action (generation) pathway 0 of 16 Mustard right of raisin box, 16 practice starts; unedited 0	Cosmos-Reason2 VLM backbone 0, flow-matching DiT action head 0 of 32 Cube then banana, 32 practice starts; unedited 9	VLM 55, action expert 20, output projection 0 of 64 Cube then banana; unedited 5; only the VLM edit transfers

	Cosmos3 Edge	**GR00T N1.7**	**π0.5**
Cube then banana	21/32 · 66%	9/32 · 28%	3/16 · 19%
Eyeglasses	2/32 · 6%	0/32 · 0%	7/16 · 44%
Cube left of bowl	13/32 · 41%	1/32 · 3%	0/16 · 0%
Yellow on red	22/32 · 69%	0/16 · 0%	9/16 · 56%
Mustard in left bin	11/32 · 34%	0/32 · 0%	3/16 · 19%
Three-cube tower	2/32 · 6%	0/16 · 0%	6/16 · 38%
Flat box	0/32 · 0%	0/32 · 0%	0/16 · 0%
Mustard right of raisin box	0/32 · 0%	0/32 · 0%	0/16 · 0%
Cube in bowl	21/32 · 66%	28/32 · 88%	16/16 · 100%

Each cell: the starts the unedited robot completed, of the starts tried, and that share; stronger shading is a higher share.

	Cosmos3 Edge	**π0.5**
Cube then banana	±0 / 32	**+38 / 64***
Eyeglasses	−3 / 16	**+9 / 32***
Cube left of bowl	**−13 / 32***	+5 / 32
Yellow on red	−2 / 32	+6 / 32
Mustard in left bin	**−9 / 32***	−1 / 32
Three-cube tower	±0 / 32	±0 / 32
Flat box	±0 / 16	±0 / 32
Mustard right of raisin box	**+56 / 64***	±0 / 32
Cube in bowl	+4 / 32	±0 / 32

gain, p < 0.05 · loss, p < 0.05 · no detectable change

Cell: change in starts completed / starts tried. * p < 0.05; outlined: the edit's own task

Each model's main edit, unchanged, on every suite task: the change in starts completed, edited minus unedited, from the same starts. Teal: a gain; near-black (near-white in dark mode): a loss (exact McNemar p < 0.05); grey: no detectable change; the outlined cell is the task the edit was built for.

Live page with every table, figure and record: ali-adeeb.com/weightsmith/cosmos · Paper: ali-adeeb.com/weightsmith/paper.pdf · Code and records: github.com/adeeb10abbas/weightsmith

Draft for discussion. Simulation only (RoboLab, Isaac Sim), one released checkpoint per architecture, shared slots 30000–30031 and 96000–96063 with identical initial scenes; π0.5 cells from the paper's evaluation name their own slots. Exact McNemar tests on paired starts.